

TrES-2b

R-0148

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-2_180706 · created 2026-07-16T08:53:56+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2458305.91471 +/- 0.00061 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.108 +/- 0.04
Transit depth (Rp/Rs)^2	1.16 +/- 0.87 [%]
Orbital Inclination (inc)	80.11 +/- 2.83
Airmass coefficient 1 (a1)	1.046 +/- 0.037
Airmass coefficient 2 (a2)	-0.039 +/- 0.047
Scatter in the residuals of the lightcurve fit is	3.53 %
Best Comparison Star	#3 - [364, 241]
Optimal Aperture	3.33
Optimal Annulus	12.27
Transit Duration (day)	0.015 +/- 0.028

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.108 $\pm$ 0.04 vs 0.1254 (deviation 0.38 σ)
Duration fit vs published	0.015 d vs 0.0746 d (deviation 1.84 σ)
Mid-time O-C	-0.5 min
Depth SNR	2.3
Residual scatter	3.53 %

Light curve

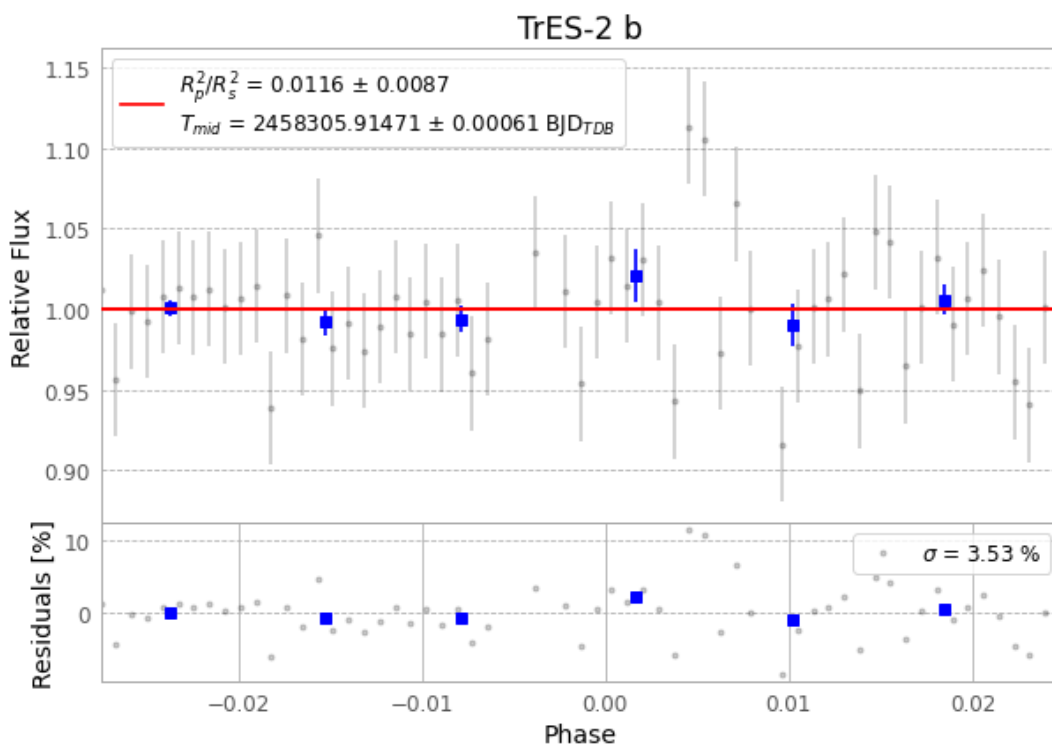


Figure 1 — FinalLightCurve_TrES-2 b_2018-07-06.png (SHA-256 48671db98153e9ea...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [357, 307], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 9f8f924abc72e433...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 3ebc8ba

Data provenance

63 FITS frames (41 MB), TRES-2180706081512.FITS → TRES-2180706112112.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-2-180706.log (ce364fedbb2c...), FinalLightCurve_TrES-2 b_2018-07-06.png (48671db98153...), FinalLightCurve_TrES-2 b_2018-07-06.csv (7bcf643d49b8...)

Compute resources

Wall time 260.5 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-16 08:53Z.